

Chapter 3: Whole numbers

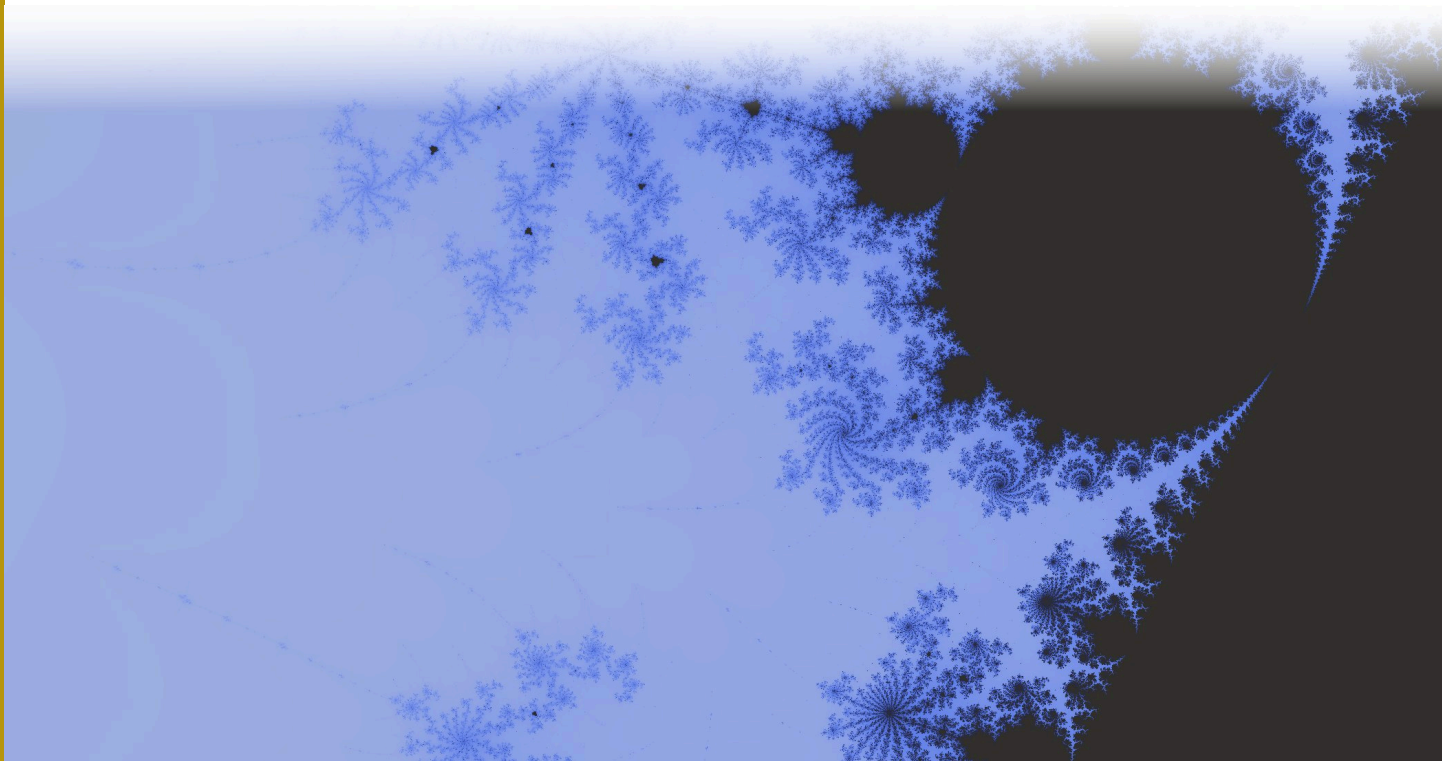
Drill practice and mixed revision of all 12 lessons · 120 questions

NSW syllabus outcome

Stage 4 Number and algebra: MA4-CAL-C-01 compares, orders and calculates with integers, and MA4-IND-C-01 (index notation, primes, factors and multiples).

CHAPTER

3



Homework. Part 1 is **drill practice**: quick questions on the main skill of each lesson in Chapter 3. Write each answer on its line. Part 2 is **mixed revision**, from **Easy** to **Challenging**. Each question is tagged with its lesson, so you can look back at that lesson's summary if you get stuck.

YOU DO On your own: Part 1, drill practice.

QUESTIONS 1–35

3.01 Rounding and estimation

Easy Round to the place given.

① 39 367 (nearest 1 000)

② 5 290 (nearest 1 000)

③ 8 319 (nearest 100)

Medium Round to the first (leading) digit.

④ 823

⑤ 484

⑥ 96 003

⑦ 37 008

3.02 Multiplication

Easy Know your times tables.

⑧ 7×11

⑨ 5×8

⑩ 4×12

⑪ 12×10

Medium Evaluate mentally. Split the number: $34 \times 6 = 30 \times 6 + 4 \times 6$.

⑫ 74×7

⑬ 31×7

⑭ 95×6

⑮ 58×5

3.03 Division

Easy Know your division facts.

⑯ $18 \div 9$

⑰ $9 \div 3$

⑱ $72 \div 9$

⑲ $36 \div 4$

Medium Evaluate mentally.

⑳ $185 \div 5$

㉑ $240 \div 4$

㉒ $208 \div 4$

㉓ $216 \div 6$

3.04 Divisibility tests

Easy Is the number divisible by the number in brackets? Write yes or no.

㉔ 9 856 (10)

㉕ 998 (10)

㉖ 7 305 (10)

㉗ 4 084 (5)

Medium Divisible by 4? Check the last two digits. Write yes or no.

㉘ 8 500

㉙ 6 067

㉚ 3 356

㉛ 8 959

3.05 Long division

Easy Know your multiples.

㉜ 17×2

㉝ 16×8

㉞ 22×6

㉟ 14×8

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 36–74

3.06 Roman numerals

Easy Write each Roman numeral as a number.

(36) LXVIII

(37) VI

(38) XL

(39) XII

Medium Write each Roman numeral as a number.

(40) CCLXXXII

(41) CMLII

(42) CLX

(43) CXIV

3.07 Index notation and powers

Easy Write in index notation.

(44) $3 \times 3 \times 3 \times 3 \times 3$

(45) 5×5

(46) $3 \times 3 \times 3$

Medium Write the base and the index, then evaluate.

(47) 4^3

(48) 3^4

(49) 2^3

(50) 5^2

3.08 Square roots and cube roots

Easy Evaluate each square root.

(51) $\sqrt{36}$

(52) $\sqrt{9}$

(53) $\sqrt{289}$

(54) $\sqrt{196}$

Medium Evaluate each cube root or cube.

(55) $\sqrt[3]{1000}$

(56) $\sqrt[3]{216}$

(57) 5^3

(58) $\sqrt[3]{125}$

3.09 Prime and composite numbers

Easy Prime (P) or composite (C)?

(59) 43

(60) 38

(61) 80

(62) 92

Medium How many factors does the number have?

(63) 75

(64) 98

(65) 32

(66) 49

3.10 Prime factors

Easy Write as a product of two prime numbers.

(67) 10

(68) 14

(69) 121

(70) 15

Medium Evaluate.

(71) 2

(72) $2^3 \times 5$

(73) 2^2

(74) 2×3

YOU DO

On your own: Part 1, drill practice.

QUESTIONS 75–90

3.11 Highest common factor

Easy Find the highest common factor (HCF).

75 16 and 40

76 24 and 20

77 3 and 18

78 15 and 10

Medium Find the HCF.

79 36 and 48

80 30 and 45

81 72 and 54

82 60 and 30

3.12 Lowest common multiple

Easy Find the lowest common multiple (LCM).

83 11 and 5

84 10 and 3

85 5 and 8

86 7 and 4

Medium Find the LCM.

87 24 and 15

88 19 and 16

89 9 and 21

90 11 and 16

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 91–100

91

Easy

Round to the nearest hundred.

4 862 Lesson 3.01

92

Easy

Evaluate.

$\sqrt{169}$ Lesson 3.08

93

Easy

Evaluate mentally.

$144 \div 6$ Lesson 3.03

94

Easy

Write as a product of two primes.

55 Lesson 3.10

95

Easy

Know your multiples.

16×6 Lesson 3.05

96

Easy

Find the LCM.

10 and 15 Lesson 3.12

97

Easy

Evaluate the power.

2^6 Lesson 3.07

98

Easy

Evaluate mentally.

15×4 Lesson 3.02

99

Easy

Prime or composite?

37 Lesson 3.09

100

Easy

Find the digit sum. Is it divisible by 3?

2 541 Lesson 3.04

YOU DO

On your own: Part 2, mixed revision.

QUESTIONS 101–110

101

Easy

Find the HCF.

16 and 40 Lesson 3.11

102

Easy

Write in Roman numerals.

86 Lesson 3.06

103

Medium

Estimate $512 + 289 - 97$. Lesson 3.01

104

Medium

Evaluate $\sqrt[3]{125}$. Lesson 3.08

105

Medium

Evaluate $738 \div 6$. Lesson 3.03

106

Medium

Write 84 as a product of primes in index notation. Lesson 3.10

107

Medium

Evaluate $1728 \div 16$. Lesson 3.05

108

Medium

Find the LCM of 12 and 16. Lesson 3.12

109

Medium

Evaluate 4^3 . Lesson 3.07

110

Medium

Evaluate 243×6 . Lesson 3.02

YOU DO On your own: Part 2, mixed revision.

QUESTIONS 111–120

111 **Medium** List the factors of 28. Lesson 3.09

112 **Medium** Is 4815 divisible by 9? Lesson 3.04

113 **Medium** Find the HCF of 42 and 70. Lesson 3.11

114 **Medium** Write 39 in Roman numerals. Lesson 3.06

115 **Challenging** Estimate $4180 \div 21$. Lesson 3.01

116 **Challenging** Evaluate $1250 \div 8$, with remainder. Lesson 3.03

117 **Challenging** Evaluate $4000 \div 27$ with remainder. Lesson 3.05

118 **Challenging** Evaluate $3^2 \times 2^4$. Lesson 3.07

119 **Challenging** List the primes between 30 and 45. Lesson 3.09

120 **Challenging** Use prime factors to find the HCF of 168 and 252. Lesson 3.11

YOU DO

On your own: extra working space. Write the question number next to your working.